A CASE-STUDY BASED REPORT

ON

Fake Product Review Detection Using NLP and Random Forest

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by

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Abstract

Fake product reviews mislead customers, damage trust in online marketplaces and unfairly reward dishonest sellers. This case study develops a supervised machine-learning model that classifies a product review as Genuine or Fake. A labelled dataset of 500 reviews with 18 columns was explored to understand class balance, data quality and the characteristics that separate the two classes. Thirteen numeric features covering rating, verified-purchase status, helpful votes, text statistics, sentiment ratios and duplicate counts were used to train a Random Forest classifier (200 trees) on an 80/20 stratified split. The model achieved 100% accuracy, precision, recall, F1 score and AUC on the 100 held-out reviews, with exclamation count, sentiment score and positive-word ratio as the most important features. Because the sample is small and highly regular, the study also discusses why these perfect scores should be interpreted with caution and how the work can be strengthened with cross-validation, larger data and text-based features.

Keywords: fake review detection, Random Forest, binary classification, exploratory data analysis, feature importance.

Introduction

Background

Online shopping platforms display customer reviews and star ratings next to every product, and shoppers use them as a trusted shortcut when deciding what to buy. Because reviews influence sales so strongly, some sellers and agencies write or buy fabricated reviews to inflate their own products or damage competitors. Such reviews are often short, repetitive, extremely enthusiastic and posted by accounts that never bought the product. Identifying them is a natural classification task for machine learning, because the patterns that distinguish fake from genuine reviews can be learned from labelled examples.

Motivation for the Project

Fake reviews mislead customers, erode trust in e-commerce platforms and penalise honest sellers. The volume of reviews posted every day is far too large for manual moderation, so platforms need automated tools that can score each review and direct human moderators to the most suspicious ones. This project was motivated by the need for a simple, explainable and data-driven detector that shows which review characteristics give fake reviews away.

Overview of the Work Carried Out

The work covers the full machine-learning workflow. A 500-review labelled dataset was loaded and inspected, exploratory data analysis was carried out, categorical fields were encoded, and 13 numeric features were selected. The data was split 80/20 with stratification, a Random Forest classifier was trained, and its predictions were evaluated with accuracy, precision, recall, F1

score, ROC-AUC, PR-AUC and a confusion matrix. Feature importance was then used to interpret the model, and the trained model was saved so that new reviews can be scored.

Scope and Responsible-Use Statement

The study is limited to a 500-row sample and 13 engineered numeric features derived from review metadata and simple text statistics. It is an academic exercise, not a production moderation system. Predictions should be used only as a flag for human review, never as the sole basis for removing a review, penalising a seller or accusing a reviewer. Every flagged review should be checked by a person, and the model must be re-validated on realistic data before any real-world use.

Problem Statement

Case-Study Scenario

An e-commerce marketplace lists 100 products and receives hundreds of reviews every month. The trust-and-safety team notices clusters of five-star reviews that repeat the same wording, use many exclamation marks and come from non-verified buyers. The team wants a lightweight model that scores each incoming review so that analysts can prioritise which ones to inspect first.

Central Case-Study Question

Can a Random Forest classifier, using only review metadata and simple text statistics, reliably distinguish fake product reviews from genuine ones, and which features drive that decision?

Formal Problem Definition

Given a feature vector $x \in \mathbb{R}^{13}$ for each review (rating, verified-purchase flag, helpful votes, word count, character count, average word length, exclamation count, question count, capital-letter ratio, positive-word ratio, negative-word ratio, sentiment score and duplicate count), learn a function $f(x) \rightarrow y$, where $y \in \{0 = \text{Genuine}, 1 = \text{Fake}\}$. This is a binary classification problem. The model is trained on labelled examples by minimising classification error, and its performance is judged on held-out data that it has not seen during training.

Objectives that Guide the Study

- Understand the structure, quality and class balance of the review dataset.
- Identify which review characteristics differ most between fake and genuine reviews.
- Encode categorical fields and select a clean set of numeric features.
- Train a Random Forest classifier on a stratified training split.
- Evaluate it with accuracy, precision, recall, F1, ROC-AUC, PR-AUC and a confusion matrix.
- Interpret the results through feature importance and discuss how trustworthy they are.

Assumptions and Limitations

- The labels (Genuine / Fake) in the dataset are assumed to be correct.
- Only 500 reviews are used, so conclusions may not generalise to a full marketplace.
- The review text is highly repetitive (only 93 unique texts across 500 reviews), which suggests the data is partly synthetic or template-generated.
- Several features are constant or redundant, which limits what the model can reveal.
- Real fake-review authors adapt, so a model trained on today's patterns can be evaded later.

Dataset and Exploratory Data Analysis

About the Dataset

The file Fake_Product_Review_Detection_RF_Dataset_Sample_500.csv contains **500 rows and 18 columns** with no missing values. The reviews span 3 January 2025 to 30 December 2025 and cover 100 distinct products. The columns are grouped below.

Column group	Columns
Identifiers	review_id, product_id
Raw content	review_text, review_date
Metadata	rating (1–5), verified_purchase (Yes/No), helpful_votes
Text statistics	word_count, char_count, avg_word_length, exclamation_count, question_count, capital_ratio
Sentiment features	positive_word_ratio, negative_word_ratio, sentiment_score
Duplication	duplicate_count
Target	label (Genuine / Fake)

Purpose of EDA

Exploratory data analysis checks data quality, reveals class balance and shows which variables separate fake from genuine reviews before any model is trained. It also exposes problems such as constant columns, redundant features and possible leakage that would otherwise make the model results look better than they really are.

Preparation Steps Actually Performed

1. Loaded the CSV with pandas and confirmed the shape (500 × 18).
2. Checked data types, missing values (none) and summary statistics.
3. Inspected unique values: verified_purchase = {Yes, No}; label = {Genuine, Fake}; rating = {1, 2, 3, 4, 5}.
4. Encoded verified_purchase (Yes = 1, No = 0) and label (Genuine = 0, Fake = 1).

5. Selected 13 numeric features as X and the encoded label as y.
6. Split the data 80/20 with stratification (random_state = 42), giving 400 training rows and 100 test rows.

The identifier, date and raw text columns were not used as model inputs.

EDA Exploration - Implementation

Class distribution. The two classes are close to balanced, so no resampling was needed.

Class	Count	Share
Genuine	262	52.4%
Fake	238	47.6%

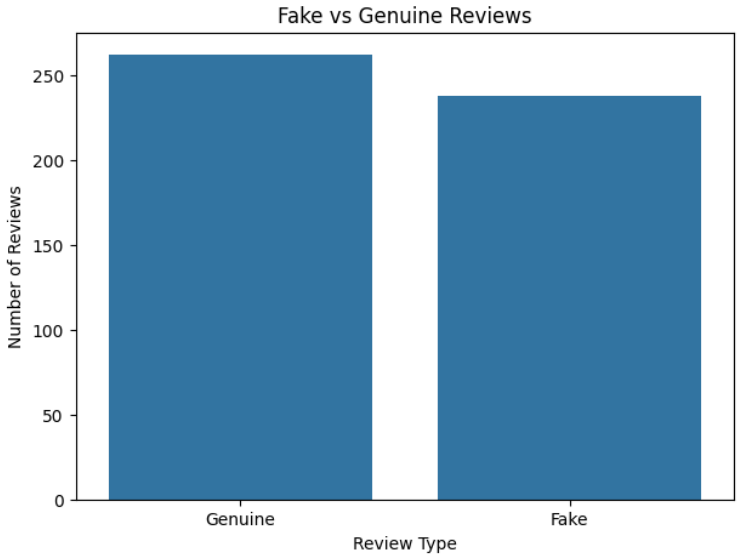


Figure 1: Fake vs. genuine reviews

Rating distribution. Ratings are skewed high: 5★ = 261, 4★ = 168, 3★ = 49, 2★ = 17 and 1★ = 5 (mean 4.33, standard deviation 0.86). Every 1★ and 2★ review is Genuine, while Fake reviews appear only at 3★ (7), 4★ (42) and 5★ (189).

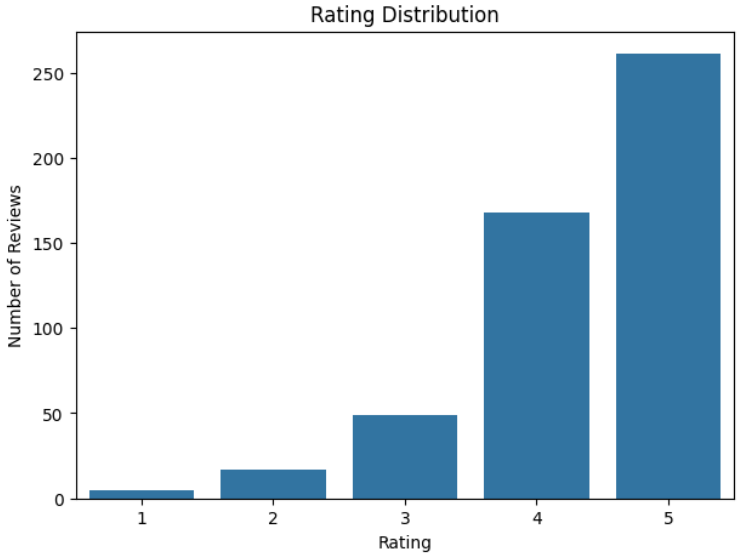


Figure 2: Rating distribution

Verified purchase vs. label. About 90% of fake reviews come from non-verified purchases, while about 81% of genuine reviews are verified.

	Fake	Genuine
Not verified (264)	213	51
Verified (236)	25	211

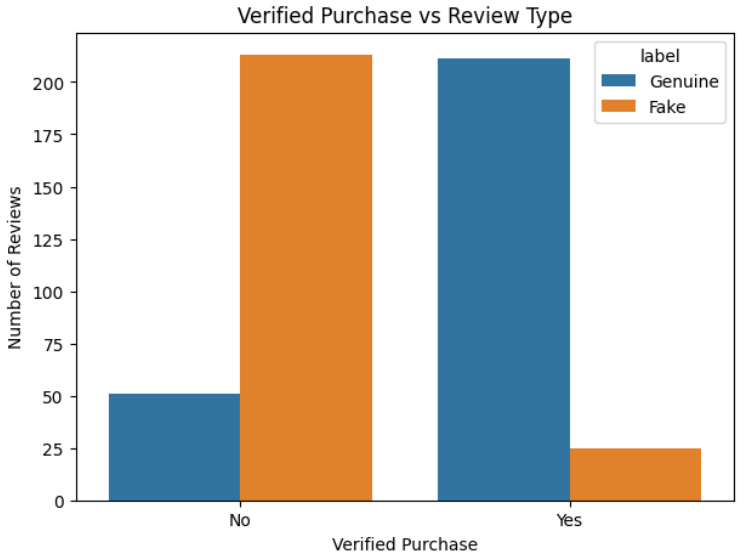


Figure 3: Verified purchase vs. review type

Feature averages by class. Fake reviews are shorter, use longer words, use more capital letters and exclamation marks, contain far more positive words, repeat verbatim and receive few helpful votes. For example, the fake text “Fantastic product!!! Super amazing!!! Five stars!!! Highly recommended!!!” appears 20 times.

Feature	Genuine	Fake
helpful_votes	52.8	7.7
word_count	14.2	9.2
avg_word_length	4.43	5.63
exclamation_count	0.00	9.64
capital_ratio	0.027	0.100
positive_word_ratio	0.030	0.309
sentiment_score	0.030	0.309
duplicate_count	0.37	5.48

Correlation analysis. The correlation matrix of the 13 features and the label shows that exclamation count, positive-word ratio, sentiment score, capital ratio and duplicate count are the strongest positive correlates of the Fake label, while helpful votes and verified purchase are negative correlates.

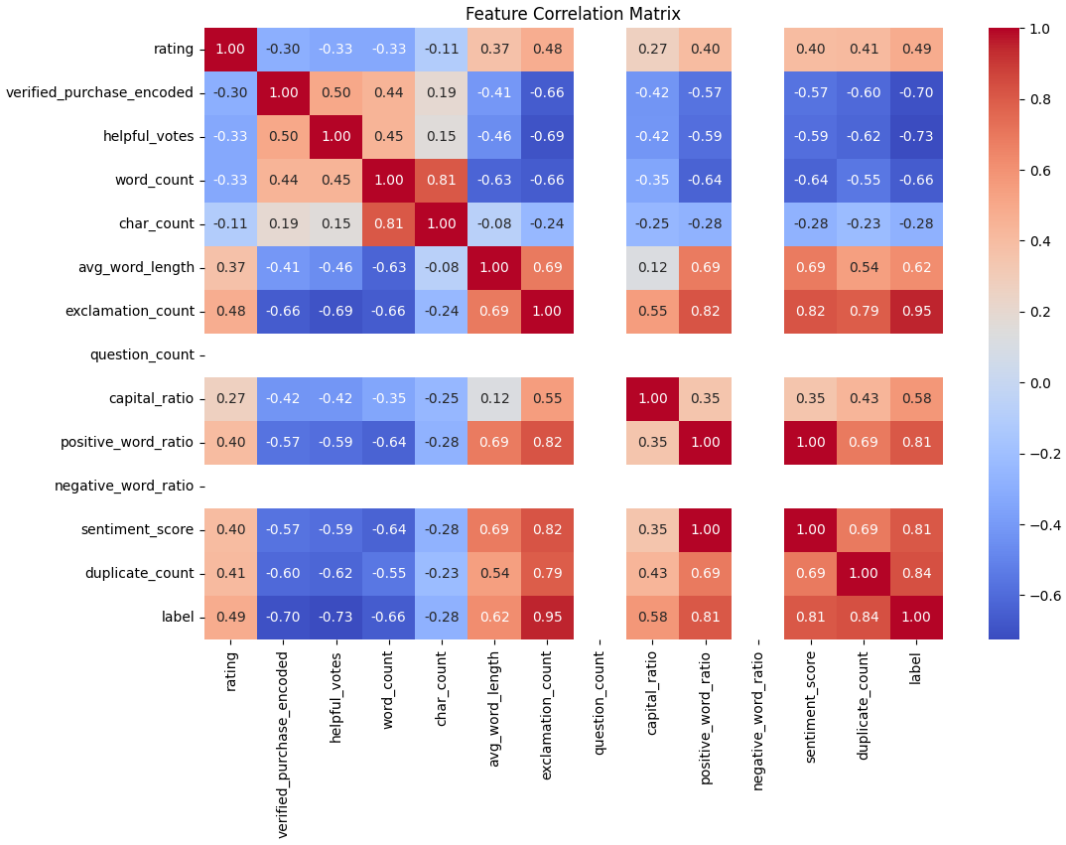


Figure 4: Feature correlation matrix

Data-quality observations.

- question_count and negative_word_ratio are 0 in every row, so they carry no information.
- sentiment_score is identical to positive_word_ratio in this sample, so the two columns are redundant.
- Every fake review has duplicate_count above 0, compared with about 27.5% of genuine reviews.
- Only 93 unique review texts exist among the 500 rows, so the same sentences recur across different reviews.

Summary of EDA Findings

The dataset is clean and nearly balanced, with no missing values. Fake reviews are consistently over-enthusiastic, repetitive, unverified and unhelpful, which makes them easy to describe statistically. At the same time, some features are constant or duplicated, and exclamation count alone separates the two classes almost perfectly (0 for every genuine review and 9.6 on average for fake ones). This foreshadows very high model scores and signals that the task, as posed by this sample, is easy.

Algorithm Selection and Implementation

Why Random Forest

Random Forest is an ensemble of decision trees. Each tree is trained on a bootstrap sample of the data using a random subset of features, and the final class is decided by majority vote. It was chosen because it handles mixed numeric features without scaling, captures non-linear interactions, resists overfitting better than a single tree, tolerates redundant features and provides feature-importance scores that make the result explainable.

Implementation

Setting	Value
Library	scikit-learn RandomForestClassifier
n_estimators	200
max_depth	20
min_samples_split	5
min_samples_leaf	2
class_weight	balanced
random_state	42
Train / test split	80 / 20, stratified (400 / 100 rows)

The model was fitted on the training set, used to predict labels and fake-class probabilities on the test set, and then evaluated. The trained model and encoders were saved, and a simple input routine lets a user enter the feature values of a new review to obtain a prediction.

Results

Metric	Score
Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1 score	1.0000
ROC-AUC	1.0000
PR-AUC	1.0000

The confusion matrix on the 100 test reviews shows 52 genuine and 48 fake reviews all classified correctly, with no false positives and no false negatives.

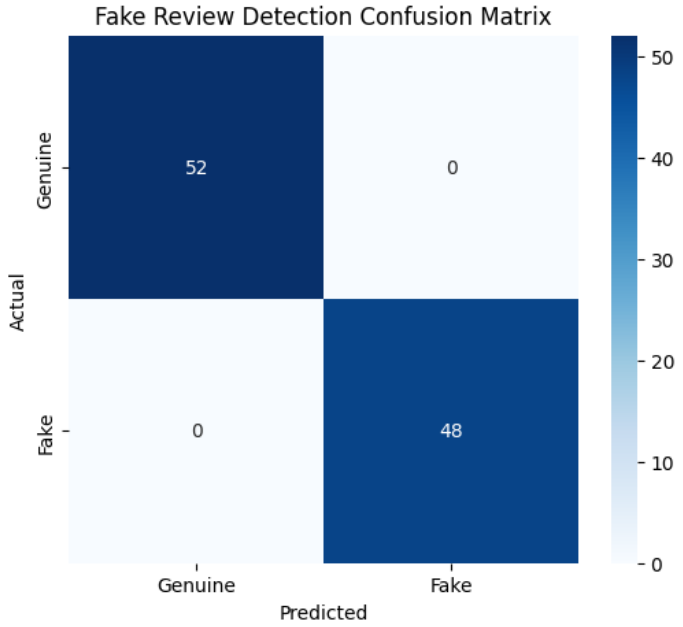


Figure 5: Confusion matrix on the test set

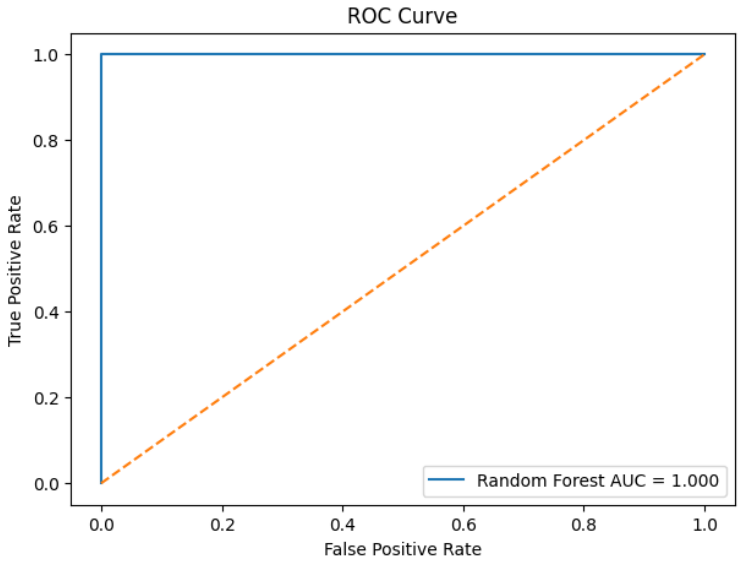


Figure 6: ROC curve (AUC = 1.0)

Feature importance. Exclamation count is by far the most influential feature, followed by the sentiment features, capital ratio and duplicate count. question_count and negative_word_ratio contribute nothing.

Rank	Feature	Importance
1	exclamation_count	0.331
2	sentiment_score	0.154
3	positive_word_ratio	0.152
4	capital_ratio	0.106
5	duplicate_count	0.093
6	helpful_votes	0.082
7	word_count	0.040
8	verified_purchase_encoded	0.019

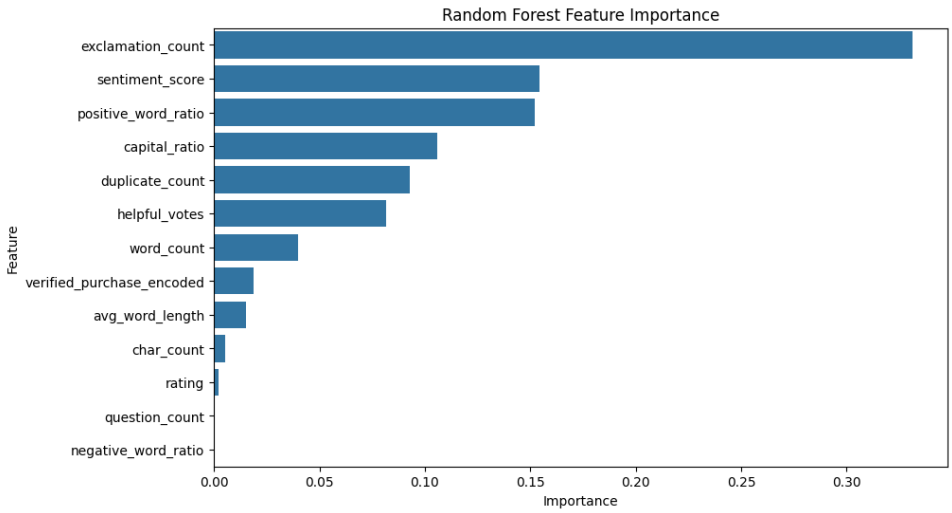


Figure 7: Random Forest feature importance

Discussion and Critical Reflection

The model classified all 100 test reviews correctly, which answers the central question in the affirmative for this dataset. However, perfect scores should be treated with caution. The test set is small, the data is repetitive and appears synthetic, and a handful of strongly separating features (exclamation count, positive-word ratio and duplicates) make the problem nearly trivial. Real-world fake reviews are subtler, so these scores are unlikely to carry over to a live platform. Sensible next steps are k-fold cross-validation, testing on a larger and more realistic dataset, removing redundant features, comparing against baselines such as logistic regression and gradient boosting, and adding text-based features such as TF-IDF.

Conclusion

This case study showed that fake product reviews in the sample differ markedly from genuine ones: they are unverified, repetitive, exclamation-heavy, strongly positive and rarely marked helpful. A Random Forest classifier trained on 13 simple features achieved perfect accuracy, precision, recall, F1 score and AUC on the held-out test set, with exclamation count as the most influential feature. Because the sample is small and highly regular, the result demonstrates the method rather than guaranteeing real-world performance, and any deployment should keep a human reviewer in the loop.

Signature of the Team members